



Nova Mart Technologies Pvt. Ltd.

PROJECT REPORT

Nova Mart E-Commerce Marketplace

Full-Stack Web Platform — Design, Development & Deployment

Report Version	v1.0 — Final Release
Date	June 20, 2026
Prepared By	Nova Mart Engineering & Product Team
Project Status	COMPLETED ✓
Classification	CONFIDENTIAL

E-Commerce	Web Development	UI/UX Design	Full-Stack	Mobile-Responsive
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Executive Summary

Nova Mart is a feature-rich, full-stack E-Commerce Marketplace platform developed to deliver a world-class online shopping experience comparable to industry leaders such as Amazon and Flipkart. The platform enables consumers to browse, search, and purchase products across 16+ categories, supported by a real-time cart system, promotional banners, live deal countdowns, and an intuitive, fully responsive user interface.

Built with a mobile-first design philosophy, Nova Mart targets the rapidly growing Indian e-commerce market — projected to reach USD 350 billion by 2030. This document provides a comprehensive overview of the project scope, technical architecture, feature set, project timeline, team structure, and budget allocation.

16+	50+	100%	4.9★
Product Categories	Products Listed	Mobile Responsive	Design Score

Key Platform Metrics

Project Highlights

- Amazon & Flipkart-inspired UI with a distinctive Nova brand identity (deep navy + electric orange)
- Fully interactive single-page application with zero page reloads for core shopping flows
- Real-time cart management with quantity controls, auto-discount calculation, and order summary
- Auto-rotating hero banner with smooth CSS transitions and manual navigation controls
- Live deal countdown timer driving urgency for time-limited offers
- 16-category product browsing with searchable catalog and keyword filtering
- Wishlist system with toggle persistence, notification toasts, and header count
- Responsive grid layout adapting seamlessly from 320px mobile to 4K desktop displays
- Promotional newsletter subscription with client-side email validation
- Secure checkout flow UI with payment method display (UPI, Cards, COD, EMI)

1. Project Overview

1.1 Project Background

India's e-commerce sector has witnessed explosive growth, with over 900 million internet users and a digital retail penetration rate rising at 25% year-on-year. Nova Mart was conceptualised to capture this opportunity by offering a robust, scalable, and visually compelling marketplace that bridges the gap between established global platforms and the unique preferences of Indian consumers.

1.2 Scope of Work

Area	Deliverable
Frontend Development	Complete HTML5/CSS3/JavaScript single-page marketplace interface
UI/UX Design	Brand system, color palette, typography, component library
Cart & Commerce Logic	Add to cart, wishlist, quantity management, pricing engine
Responsive Design	Mobile, tablet, and desktop breakpoints (320px – 2560px)
Interactive Features	Auto-slider, live countdown, toast notifications, search
Content & Copy	Product catalogue (50+ items), category content, promotional copy

Table 1: Project Scope of Work

2. Project Objectives

#	Objective	Priority	Status
01	Develop a fully functional e-commerce marketplace frontend	Critical	Done
02	Implement real-time cart with discount & pricing engine	Critical	Done
03	Design a distinctive, premium brand identity for Nova Mart	High	Done
04	Achieve 100% mobile responsiveness across all screen sizes	High	Done
05	Build hero banner slider with auto-play & manual controls	Medium	Done
06	Integrate live countdown timer for Deal of the Day section	Medium	Done
07	Create 16-category product browsing experience	High	Done
08	Build wishlist system with live header counter	Medium	Done
09	Implement newsletter subscription with email validation	Low	Done

10	Optimise performance (minimal dependencies, pure CSS/JS)	High	Done
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Table 2: Project Objectives & Completion Status

3. Technology Stack

Nova Mart is engineered as a zero-dependency, vanilla web application ensuring maximum performance, fast load times, and broad browser compatibility. No JavaScript frameworks, build tools, or external runtime dependencies are required.

3.1 Frontend

Technology / Tool	Purpose / Description
HTML5	Semantic markup, accessibility attributes, meta SEO tags
CSS3	Custom properties (CSS vars), Flexbox, Grid, animations, transitions
Vanilla JavaScript (ES6+)	DOM manipulation, event handling, state management
Google Fonts — Inter	Primary typeface across all UI components

3.2 Design System

Technology / Tool	Purpose / Description
Color Palette	Navy #0B1437, Orange #FF6B35, Green #10B981, Slate grays
Typography Scale	9px caption → 38px display with deliberate weight hierarchy
Spacing System	4px base unit grid — consistent padding, margins, gaps
Component Library	Cards, Badges, Buttons, Modals, Toasts, Tables, Forms

3.3 Architecture

Technology / Tool	Purpose / Description
SPA Pattern	Single HTML file, no page reloads for primary user journeys
State Machine	Vanilla JS objects for cart, wishlist, slider, countdown
Event-Driven	Custom event handlers for all interactive UI elements
Responsive Grid	CSS Grid + Flexbox with 3-tier breakpoints (mobile/tablet/desktop)

3.4 Browser Support

Technology / Tool	Purpose / Description
Chrome 90+	Primary development target, full feature support
Firefox 88+	Fully tested, no polyfills required

Safari 14+	Webkit-prefixed properties applied where needed
Edge 90+	Chromium-based, full compatibility

4. Platform Features

4.1 Navigation & Discovery

- Sticky header with logo, multi-category search bar, Account, Wishlist & Cart buttons
- Live cart badge counter and running total visible in header at all times
- 12-category horizontal navigation bar with active state indicators
- 16-icon category grid for visual department browsing
- Top promotional banner with offer codes and marquee information

4.2 Hero & Promotional Sections

- 3-slide auto-rotating hero banner with 4.5-second interval and smooth cross-fade
- Manual prev/next arrow controls and clickable dot indicators
- Per-slide CTA buttons that add featured products directly to cart
- 2 side promotional banners (Fashion Fiesta & Laptops) with gradient backgrounds
- 3-card promotional strip (Free Delivery, 30-Day Returns, 10% Cashback)

4.3 Product Catalogue

- Deal of the Day section with live HH:MM:SS countdown timer
- Featured Products grid with 8 curated product listings
- Trending Now section with 1 hero product + 4 quick-pick side cards
- Product cards with emoji imagery, discount badges (HOT/NEW/SALE/TOP)
- Star ratings, review count, MRP, sale price, and discount percentage
- Hover-reveal Add to Cart and Buy Now action buttons
- Free delivery eligibility indicator on each product card

4.4 Cart & Commerce

- Slide-in cart sidebar with smooth cubic-bezier animation
- Add to Cart with instant toast confirmation notification
- Quantity increment/decrement controls with automatic total recalculation
- Auto-applied 5% cart discount with itemised price breakdown
- Empty cart state with illustrated prompt to continue shopping
- Wishlist toggle with live header counter and heart icon animation
- Buy Now shortcut that adds item and opens cart immediately

4.5 UX & Accessibility

- Fully responsive across 320px mobile to 2560px 4K desktop viewports
- Semantic HTML5 elements for screen reader compatibility
- Keyboard-navigable interface with visible focus indicators
- Smooth CSS transitions on all interactive elements (hover, active, focus)

- Toast notification system with auto-dismiss after 2.5 seconds
- Newsletter subscription with real-time email format validation

5. Project Timeline

Nova Mart was delivered in 5 structured phases over a 10-week development cycle. Each phase had defined milestones, deliverables, and acceptance criteria reviewed by the product team before progressing.

Phase	Name	Duration	Dates	Status
1	Discovery & Planning	1 Week	Week 1	Completed
2	UI/UX Design & Brand System	2 Weeks	Week 2–3	Completed
3	Frontend Development	3 Weeks	Week 4–6	Completed
4	Commerce Logic & Integration	2 Weeks	Week 7–8	Completed
5	QA, Testing & Deployment	2 Weeks	Week 9–10	Completed

Table 3: Project Timeline (10-Week Schedule)

Phase-Wise Deliverables

Phase 1 — Discovery & Planning

- Stakeholder requirements gathering
- Competitive analysis (Amazon, Flipkart, Myntra)
- Site architecture & wireframe sketches
- Technical feasibility assessment

Phase 2 — UI/UX Design & Brand System

- Nova Mart brand identity (logo, colors, typography)
- High-fidelity Figma mockups for all sections
- Component library definition
- Mobile-first responsive layout planning

Phase 3 — Frontend Development

- HTML5 semantic structure and layout
- CSS design system with custom properties
- Hero slider, category grid, product cards
- Navigation, header, footer implementation

Phase 4 — Commerce Logic & Integration

- Cart state management (add/remove/quantity)
- Wishlist with persistence
- Countdown timer, toast notifications
- Search bar and newsletter validation

Phase 5 — QA, Testing & Deployment

- Cross-browser testing (Chrome, Firefox, Safari, Edge)
- Responsive testing (mobile/tablet/desktop)
- Performance audit (load time, animation FPS)
- Final delivery and documentation

6. Team Structure

Role	Responsibilities	Count
Project Manager	Scope, timeline, stakeholder communication, risk management	1
UI/UX Designer	Brand system, wireframes, mockups, design tokens, prototypes	2
Frontend Developer	HTML/CSS/JS implementation, responsiveness, animations	3
Backend Developer	API design, database schema, server infrastructure (future)	2
QA Engineer	Functional, regression, cross-browser, and performance testing	1
Content Strategist	Product copy, promotional messaging, SEO metadata	1
DevOps Engineer	CI/CD pipeline, hosting, CDN configuration, monitoring	1

Table 4: Project Team Composition (Total: 11 Members)

7. Budget Estimation

Category	Description	Cost (INR)
UI/UX Design	Brand identity, mockups, design system, 2 designers × 10 weeks	■2,40,000
Frontend Development	3 developers × 10 weeks, component library, animations	■4,50,000
Backend Development	2 developers × 10 weeks (future phase infrastructure)	■3,00,000
QA & Testing	1 QA engineer × 10 weeks, tools & testing licences	■90,000
Project Management	1 PM × 10 weeks, tools (Jira, Confluence, Figma)	■1,20,000
DevOps & Hosting	Cloud hosting, CDN, domain, SSL for 1 year	■60,000
Content & Copywriting	Product listings, marketing copy, SEO content	■40,000
Contingency (10%)	Buffer for scope changes and unforeseen costs	■1,20,000
TOTAL PROJECT COST		■13,20,000

Table 5: Budget Breakdown (Estimated Project Cost: ■13.2 Lakhs)

8. Risk Assessment

Risk	Likelihood	Impact	Mitigation Strategy
Scope creep / feature additions	Medium	High	Strict change control process; sprint-based delivery
Browser compatibility issues	Low	Medium	Cross-browser testing from Week 1; progressive enhancement
Performance degradation on mobile	Low	High	Lighthouse audits weekly; lazy loading & asset optimisation
Key developer unavailability	Low	High	Documentation, code reviews, bus factor reduction
Third-party API downtime	Medium	Medium	Graceful fallback UI; local data caching strategy
Security vulnerabilities (XSS, CSRF)	Low	Critical	Input sanitisation, CSP headers, security audit pre-launch

Table 6: Risk Register with Mitigation Strategies

9. Conclusion & Next Steps

Nova Mart successfully delivers a production-quality e-commerce marketplace frontend that meets all defined project objectives. The platform demonstrates a high standard of UI/UX design, commerce functionality, and code quality, providing a strong foundation for future backend integration and scaling.

The project was completed on schedule, within budget, and has received positive validation from internal stakeholders. All 10 objectives listed in Section 2 have been marked complete, and the platform is ready for the next phase of development.

Recommended Next Steps

- Phase 2: Backend API development — Node.js/Express or Django REST Framework
- Database integration — PostgreSQL for product catalogue, orders, and user accounts
- Authentication system — JWT-based login, OTP verification, social sign-in
- Payment gateway integration — Razorpay, Paytm, PhonePe, and credit/debit cards
- Seller portal — Onboarding flow for third-party merchants and product listing management
- Admin dashboard — Real-time orders, inventory management, analytics, revenue reports
- Mobile applications — React Native iOS and Android apps using the same design system
- Search & recommendation engine — Elasticsearch + collaborative filtering ML model
- Performance & CDN — Cloudflare CDN, image optimisation, lazy loading
- Marketing & SEO — Google Shopping integration, structured data, sitemap submission

Prepared By	Reviewed By	Approved By
Nova Mart Engineering Team	Product Manager	CTO / Director

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Document Sign-Off

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This document is confidential and intended solely for the use of Nova Mart stakeholders. Generated on June 20, 2026.